

VoiceDesk

AI Voice Agent Platform for Indian Businesses

System Design Document

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Apprend Technologies

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1. Executive Summary

VoiceDesk is an AI-powered voice agent platform purpose-built for Indian small and medium businesses. The platform enables businesses to automate phone-based customer interactions across Hindi and five major regional languages: Tamil, Telugu, Marathi, Bengali, and Kannada.

The platform addresses a critical gap in the Indian market where businesses rely heavily on phone communication but lack affordable automation tools. VoiceDesk handles five key use cases: appointment booking, payment reminders, lead qualification, order status inquiries, and general customer support.

By combining state-of-the-art Automatic Speech Recognition (ASR), Large Language Models (LLMs), and Text-to-Speech (TTS) engines, VoiceDesk delivers natural, context-aware conversations that adapt to the caller's language and intent in real-time.

Target Market: Indian SMBs across healthcare (clinics, diagnostic centers), e-commerce (D2C brands), financial services (microfinance, insurance agents), education (coaching institutes), and professional services (law firms, CA firms).

Key Differentiator: Native multilingual support for Indian languages with cultural context awareness, competitive per-minute pricing, and seamless WhatsApp handoff when voice interaction is insufficient.

2. System Architecture

2.1 Technology Stack

Layer	Technology	Purpose
Runtime	Python 3.11+	Backend services and AI pipeline
Backend Framework	FastAPI	REST API and WebSocket endpoints
Frontend	React / Next.js 14	Admin dashboard and agent builder
Primary Database	PostgreSQL 16	Transactional data, call metadata
Cache / Queue	Redis 7 + BullMQ	Job queues, session cache, pub/sub
Vector Store	Qdrant	Conversation embeddings for context retrieval
AI Gateway	OpenRouter	LLM routing (free-tier models)
ASR Engine	Whisper / Deepgram	Speech-to-text with Hindi/regional support
TTS Engine	ElevenLabs / Azure	Natural voice synthesis in Indian languages
Telephony	Exotel / Knowlarity	Indian telephone network integration
Object Storage	S3-compatible (MinIO)	Call recordings and media files
Real-time	WebSocket + WebRTC	Live call streaming and monitoring

2.2 Voice Pipeline Architecture

The voice pipeline processes calls through a sequential chain of services, each optimized for low-latency operation:

Inbound Call Flow

1. Caller dials a VoiceDesk-assigned phone number via Exotel/Knowlarity.
2. Telephony webhook triggers the Call Router, which identifies the business and assigned voice agent.
3. Audio stream is established via WebRTC for real-time bidirectional communication.
4. ASR engine (Whisper/Deepgram) transcribes caller speech, detecting language automatically.
5. Transcribed text is sent to the LLM (via OpenRouter) with conversation context from Qdrant.
6. LLM generates a response following the configured conversation flow and business rules.
7. TTS engine converts the response to speech in the detected language.
8. Audio response is streamed back to the caller in real-time.

Outbound Call Flow

1. Business schedules outbound calls via the dashboard (e.g., payment reminders, appointment confirmations).
2. BullMQ job scheduler picks up the call at the scheduled time.
3. Telephony provider initiates the call; DND registry is checked before dialing.

4. Once connected, the same ASR-LLM-TTS pipeline handles the conversation.
5. Call outcome is logged and triggers configured follow-up actions (e.g., send WhatsApp confirmation).

2.3 Real-time Streaming Architecture

VoiceDesk employs a WebRTC-based streaming architecture for sub-200ms end-to-end latency. The system uses a media server to bridge between the PSTN telephony network and the WebRTC pipeline, enabling real-time call monitoring from the dashboard.

Key performance targets: ASR latency under 300ms, LLM inference under 500ms, TTS synthesis under 200ms, with total round-trip under 1.5 seconds for a natural conversational experience.

2.4 Call Recording Storage

All call recordings are stored in S3-compatible storage (MinIO for self-hosted, AWS S3 for cloud). Recordings are encrypted at rest using AES-256. Retention policies are configurable per business (default: 90 days). Recordings are stored in Opus format for optimal compression while maintaining audio quality suitable for compliance review.

3. Data Model

3.1 Core Tables

Table	Description	Key Fields
businesses	Tenant registration and config	id, name, phone, plan, settings_json, created_at
voice_agents	AI agent personas and behavior	id, business_id, name, language, voice_id, persona, flow_json
phone_numbers	Assigned telephony numbers	id, business_id, agent_id, number, provider, region, status
call_logs	Individual call records	id, business_id, agent_id, caller_number, duration_sec, status, sentiment
conversations	Turn-by-turn dialogue	id, call_id, role, content, language, confidence, timestamp
intents	Detected caller intents	id, business_id, name, description, action_type, parameters_json
call_recordings	Audio file references	id, call_id, storage_path, duration_sec, size_bytes, encrypted
analytics	Aggregated metrics	id, business_id, date, total_calls, avg_duration, resolution_rate
billing_usage	Per-minute billing records	id, business_id, month, minutes_used, amount_paise, plan_id

All tables include `business_id` for multi-tenant isolation. PostgreSQL Row-Level Security (RLS) is enabled as a backstop. Soft delete is enforced via `deleted_at` timestamps.

4. Core Features

4.1 Voice Agent Builder

A drag-and-drop visual editor for designing conversation flows. Business owners can create multi-step call scripts with branching logic, intent detection nodes, API call nodes (for CRM/calendar lookups), and handoff nodes. Flows are stored as JSON and interpreted by the conversation engine at runtime.

4.2 Multilingual Support

Language	ASR Support	TTS Support	Market Coverage
Hindi	Whisper + Deepgram	ElevenLabs + Azure	~40% of Indian population
English	Whisper + Deepgram	ElevenLabs + Azure	Business communication
Tamil	Whisper	Azure Neural	Tamil Nadu, Sri Lanka
Telugu	Whisper	Azure Neural	Andhra Pradesh, Telangana
Marathi	Whisper	Azure Neural	Maharashtra
Bengali	Whisper	Azure Neural	West Bengal, Bangladesh
Kannada	Whisper	Azure Neural	Karnataka

Language detection is automatic. The system identifies the caller's language within the first 3-5 seconds and switches the agent's response language accordingly. Code-switching (mixing Hindi and English) is supported natively.

4.3 Real-time Call Monitoring Dashboard

Supervisors can monitor live calls in real-time, viewing the transcription as it happens, caller sentiment indicators, and agent confidence scores. One-click takeover allows a human agent to seamlessly join or replace the AI agent mid-call.

4.4 Call Analytics and Sentiment Analysis

Post-call analytics include sentiment trajectory (positive/negative/neutral over the call duration), intent classification accuracy, resolution status, and caller satisfaction prediction. Weekly and monthly reports are auto-generated and emailed to business owners.

4.5 WhatsApp Handoff

When a voice conversation cannot be resolved (e.g., the caller needs to share a document, the AI confidence drops below 70%, or the caller explicitly requests text communication), VoiceDesk initiates a seamless WhatsApp handoff. The conversation context is transferred to GoSumo's WhatsApp channel, and the caller receives a WhatsApp message with the conversation summary.

4.6 CRM Integrations

Native integrations with GoSumo and HomeNex enable the voice agent to access customer records, appointment history, and order status in real-time during calls. A generic webhook-based integration framework supports third-party CRMs.

4.7 Appointment Booking with Calendar Sync

Voice agents can check real-time availability via Google Calendar or Outlook integration, book appointments during the call, and send confirmation via SMS/WhatsApp. Rescheduling and cancellation flows are pre-built.

4.8 Payment Reminder Automation

Automated outbound calls for payment reminders with configurable escalation: first reminder (friendly), second reminder (firm), third reminder (urgent with consequences). Integration with payment gateways enables on-call payment links via SMS.

4.9 Lead Qualification Scripts

AI-driven lead scoring during calls based on configurable qualification criteria (BANT framework: Budget, Authority, Need, Timeline). Qualified leads are automatically pushed to the CRM with a score and call summary.

5. API Design

5.1 RESTful Endpoints

Method	Endpoint	Description
POST	/api/v1/agents	Create a new voice agent
GET	/api/v1/agents/:id	Retrieve agent configuration
PUT	/api/v1/agents/:id/flow	Update conversation flow
POST	/api/v1/calls/initiate	Initiate an outbound call
GET	/api/v1/calls/:id	Get call details and transcript
GET	/api/v1/calls/:id/recording	Stream call recording
POST	/api/v1/webhooks/telephony	Receive telephony events
POST	/api/v1/webhooks/call-status	Call status change events
GET	/api/v1/analytics/dashboard	Dashboard metrics summary
GET	/api/v1/analytics/calls	Call analytics with filters
POST	/api/v1/phone-numbers/provision	Provision a new phone number
GET	/api/v1/billing/usage	Current billing cycle usage

All endpoints require JWT authentication and include `business_id` extraction from the token. Rate limiting is enforced at 100 requests/minute per business. Webhook endpoints verify HMAC-SHA256 signatures before processing.

6. Revenue Model

6.1 Pricing Tiers

Plan	Monthly Fee	Per-Minute Rate	Included Minutes	Features
Starter	\$49/mo	\$0.15/min	200 min	1 agent, 1 language, basic analytics
Growth	\$99/mo	\$0.10/min	500 min	3 agents, 3 languages, CRM integration
Business	\$199/mo	\$0.05/min	2000 min	Unlimited agents, all languages, API access
Enterprise	Custom	Custom	Custom	Dedicated infrastructure, SLA, custom integrations

Overage minutes are billed at the plan's per-minute rate. All prices are in USD; INR billing is available at the prevailing exchange rate. Annual plans receive a 20% discount.

7. Deployment Architecture

7.1 Infrastructure

Component	Provider	Specification
Application Server	Hetzner VPS	4 vCPU, 16 GB RAM, 160 GB NVMe
Database	PostgreSQL 16	Managed on same VPS, daily backups
Cache/Queue	Redis 7	Managed on same VPS, persistence enabled
Reverse Proxy	Caddy	Auto TLS, HTTP/2, rate limiting
Process Manager	systemd	Auto-restart, log management
Object Storage	MinIO / S3	Call recordings, encrypted at rest
Monitoring	Grafana + Prometheus	Metrics, alerts, dashboards

7.2 Service Configuration

The API server runs as a systemd service on port 3010. The dashboard is served as a static Next.js export on port 3011. Caddy handles TLS termination and reverse proxying for both services. The WebSocket endpoint for real-time monitoring is proxied through Caddy with appropriate timeout configuration.

8. Security

8.1 Call Security

All call audio is encrypted in transit using SRTP (Secure Real-time Transport Protocol). Call recordings are encrypted at rest using AES-256. PII (phone numbers, names) is masked in application logs using a configurable masking pipeline.

8.2 Regulatory Compliance

TRAI (Telecom Regulatory Authority of India) compliance is built-in:

1. DND (Do Not Disturb) registry check before every outbound call.
2. Calling hours restricted to 9 AM - 9 PM IST as per TRAI regulations.
3. Mandatory caller ID display on all outbound calls.
4. Call recording consent announcement at the start of each call.
5. Opt-out mechanism available during every call.

8.3 Data Protection

Multi-tenant isolation via PostgreSQL RLS. API authentication via JWT with short-lived tokens (15 min) and refresh tokens (7 days). Webhook signatures verified using HMAC-SHA256. Rate limiting at API gateway and per-tenant levels.

9. Testing Strategy

9.1 Test Categories

Category	Scope	Tools	Coverage Target
Unit Tests	Service logic, utils, parsers	pytest, unittest.mock	>85% line coverage
Integration Tests	API endpoints, DB operations	pytest + testcontainers	>75% path coverage
Voice Pipeline Tests	ASR/TTS accuracy, latency	Custom test harness	>90% word accuracy
Load Tests	Concurrent call handling	Locust	100 concurrent calls
E2E Tests	Full call flows	Playwright + mock telephony	Critical paths

9.2 Mock Telephony

A mock telephony service simulates Exotel/Knowlarity APIs for development and CI/CD. It supports configurable call scenarios (pickup, no-answer, busy, network-error) and records all API interactions for assertion in tests.

10. Milestones

Phase	Timeline	Deliverables	Success Criteria
MVP	Weeks 1-4	Inbound calls, Hindi + English, basic agent builder, call logging	50 test calls with >80% intent accuracy
Alpha	Weeks 5-6	Outbound calls, payment reminders, WhatsApp handoff	10 pilot businesses onboarded
Beta	Weeks 7-8	All 7 languages, CRM integration, analytics dashboard	100 daily calls across pilots
Polish	Weeks 9-10	Call monitoring, sentiment analysis, billing system	<2s avg response latency
GA	Weeks 11-12	Production hardening, documentation, marketing site	Public launch with SLA

Each phase includes a go/no-go review with defined success criteria. The MVP focuses on proving the voice pipeline's reliability and accuracy in Hindi and English before expanding to regional languages.

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